A Survey on Hybrid Deep Learning Approaches for Robust Deepfake Detection

Navanith Krishna R, Navya Billalar, Nikhilesh C, Pooja Gaikwad
Department of Computer Science and Engineering, B.M.S. College of Engineering, Bengaluru, India

***Abstract*—Deepfakes, generated using advanced deep learning architectures such as Generative Adversarial Networks (GANs), have become a profound challenge for digital authenticity, privacy, and security. Their dual use—for creative content and malicious purposes—has created an urgent need for robust detection mechanisms. Detection systems must address not only dataset overfitting and dependence on large labeled data but also the broad spectrum of real-world distortions, including compression artifacts, noise, and adversarial manipulations. This survey adopts a workflow-oriented perspective, analyzing detection pipelines that begin with raw data acquisition and validation, followed by multi-stage preprocessing (noise addition, compression), RGB and frequency-domain analysis, and hierarchical feature extraction using both Convolutional Neural Networks (CNNs) and Transformer-based models. Detailed coverage is given to classifier design, spanning spatial, sequential, and spectral features, along with analytics and metadata tracking for comprehensive system benchmarking. Categories of surveyed models include CNN-based methods, self-attentive Transformers, frequency-domain approaches, and hybrid systems that integrate modular feature pathways, as depicted in contemporary literature and illustrated in standard system flowcharts. For each, comparative strengths, weaknesses, and results on large-scale benchmarks are reported. Research gaps are identified relating to cross-dataset generalization, computational efficiency, adversarial robustness, and ethical deployment. The survey concludes by proposing future directions, including lightweight hybrid frameworks, self-supervised domain adaptation, and rigorous evaluation protocols to ensure scalable, trustworthy deepfake detection in practice.**

***Index Terms*—Deepfake Detection, Hybrid Architectures, Convolutional Neural Networks, Transformers, Frequency-Domain Analysis, Self-Supervised Learning, Workflow Pipelines**

I Introduction

Deepfakes are artificially generated media in which a person's likeness is altered or synthesized using advanced deep learning techniques. Typically powered by Generative Adversarial Networks (GANs) and autoencoders, these models can create hyper-realistic images and videos that are increasingly difficult to distinguish from authentic content. While such technology has legitimate applications in entertainment, filmmaking, education, and assistive technologies, its misuse poses serious risks. Deepfakes have already been exploited for misinformation campaigns, identity theft, financial fraud, and reputational damage, thereby threatening individual privacy as well as societal trust in digital media.

The growing accessibility of deepfake generation tools has intensified the need for robust detection systems. However, current detection techniques face several challenges. Many approaches overfit to manipulation-specific artifacts and thus fail to generalize to unseen deepfake techniques. Others rely on large-scale annotated datasets, which are time-consuming and expensive to prepare. Furthermore, complex architectures such as Transformers achieve high accuracy but demand heavy computational resources, limiting their deployment in real-time or on mobile and resource-constrained devices. Finally, robustness remains a major concern, as performance often degrades under real-world distortions such as compression, noise, or blurring.

To address these issues, researchers have explored multiple detection strategies, ranging from Convolutional Neural Networks (CNNs) that extract spatial inconsistencies, to Transformer models that leverage long-range dependencies, to frequency-domain methods that analyze spectral cues. More recently, hybrid frameworks that integrate CNNs, Transformers, and frequency-based features have demonstrated promising results in balancing accuracy and efficiency.

This survey provides a comprehensive review of such approaches and contributes by:

- Presenting a taxonomy of deepfake detection methods based on spatial, spectral, and hybrid learning strategies.
- Comparing CNN-based, Transformer-based, frequency-domain, and hybrid models in terms of strengths, limitations, and dataset performance.
- Identifying key research gaps, particularly in generalization, robustness, computational efficiency, and scalability.
- Proposing promising directions including lightweight hybrid architectures, self-supervised learning, domain adaptation, and ethical deployment practices.

II Literature Survey

MesoNet: A Compact Facial Video Forgery Detection Network introduces two lightweight CNN architectures, Meso-4 and MesoInception-4, for efficient detection of manipulated face videos. By focusing on mesoscopic-level artifacts rather than fine-grained pixel differences, the model achieves high accuracy on FaceForensics and Deepfake datasets while maintaining computational efficiency. The compactness of MesoNet makes it suitable for real-time applications and resource-constrained devices. The study establishes that lightweight models can effectively capture manipulation artifacts, reducing reliance on heavy architectures and enabling broader applicability of forgery detection methods in practical settings [1].

Watch Your Up-Convolution: CNN-Based Fake Image Detection explores how generative models using up-sampling layers leave distinct spectral artifacts in fake images. By analyzing

the Fourier domain, the authors demonstrate that CNN-based detectors trained on spectral representations achieve superior generalization across different GAN architectures. Unlike spatial-domain models that overfit to dataset-specific artifacts, this frequency-domain approach captures universal generation inconsistencies. Extensive experiments confirm robustness against unseen manipulations. The study provides theoretical insights into the roots of detection success, emphasizing that spectrum-level abnormalities are fundamental cues for identifying synthetic content [2].

The paper F2Trans: High-Frequency Fine-Grained Transformer for Face Forgery Detection introduces a transformer-based architecture that integrates a high-frequency filter bank with a multi-head self-attention mechanism. The model decomposes input images into frequency bands using wavelet transforms, emphasizing high-frequency sub-bands (LH, HL, HH) where manipulation artifacts are most prominent. These filtered representations are tokenized and processed by a Central Difference Attention Transformer (CDAT) block, which models both local and global dependencies. The architecture also includes a High-frequency Wavelet Sampler (HWS) and Locality Skip Connection (LSC) to enhance spatial awareness and fine-grained artifact detection. Trained on datasets like FaceForensics++, F2Trans achieves state-of-the-art accuracy, particularly in cross-dataset and robustness tests, by focusing on subtle texture inconsistencies and frequency domain anomalies that evade traditional CNN-based detectors [3].

Hierarchical Frequency-Assisted Interactive Networks for Face Manipulation Detection proposes HFI-Net, a model that integrates spatial and frequency features through hierarchical representation and interactive fusion modules. Recognizing that manipulations distort both spatial textures and frequency spectra, HFI-Net learns multi-scale features to enhance robustness. The network outperforms state-of-the-art detectors on datasets such as FaceForensics++ and Celeb-DF, particularly under cross-dataset evaluation. By explicitly combining spectral and spatial cues, the framework achieves better generalization and resilience to unseen manipulations, underscoring the importance of multi-domain feature integration for reliable deepfake detection [4].

Detecting Deepfakes with Self-Blended Images proposes a data augmentation pipeline that generates self-blended images by alpha-compositing pairs of genuine and manipulated images. The blending simulates varying levels of manipulation intensity, creating a diverse training set with a spectrum of artifact granularities. The augmented dataset is used to train CNN-based detectors, such as XceptionNet, making them more sensitive to subtle manipulations. The model is evaluated on standard benchmarks, demonstrating improved robustness and accuracy in detecting deepfakes with low-level artifacts. The technical contribution is the use of image blending to simulate real-world forgery scenarios, enhancing the generalization of deepfake detectors [5].

End-to-End Reconstruction-Classification Learning for Face Forgery Detection presents a dual-task framework combining an autoencoder and a binary classifier. The encoder-decoder reconstructs authentic face images, optimized with pixel-wise L2 loss, while a shared feature extractor feeds into a classifier trained with binary cross-entropy. This joint training regularizes the feature space, forcing the network to learn intrinsic facial features and suppress manipulation artifacts. The classifier leverages these robust features to distinguish real from fake images. The model is trained end-to-end, with adversarial and reconstruction losses constraining the latent space, resulting in improved generalization to unseen forgeries and minimization of artifacts while preserving identity-specific features [6].

DomainForensics: Exposing Face Forgery Across Domains via Bi-Directional Adaptation addresses the domain shift problem in deepfake detection using a domain adversarial neural network (DANN) framework. The architecture includes gradient reversal layers and adversarial discriminators to align feature distributions between source (labeled) and target (unlabeled) domains. The model employs cycle-consistent style transfer at both image and feature levels, ensuring that the feature extractor learns domain-invariant representations. Cross-domain cycle consistency is enforced using a cycle loss, and adversarial training is used to minimize domain discrepancy. The method is evaluated on multiple datasets, showing significant improvements in cross-dataset generalization and robustness against domain-specific artifacts [7].

Boosting Deepfake Feature Extractors Using Unsupervised Domain Adaptation presents a multi-stage unsupervised domain adaptation framework that uses diffusion models for style transfer and prototypical contrastive loss for feature alignment. The architecture generates stylized images for both source and target domains, extending the feature representation space. Feature-level alignment is achieved using prototypical consistency guided supervision, and an adversarial discriminator enforces style invariance. The method is applied as a feature booster to pre-trained models like Xception, ResNet-50, EfficientNet-B4, and Swin-T. Cross-dataset evaluations show significant improvements in AUC scores, demonstrating the effectiveness of the domain adaptation strategy in enhancing generalization without domain-specific fine-tuning [6].

Deepfake Forensics via an Adversarial Game employs a GAN-based adversarial training setup, where a generator creates adversarial perturbations to camouflage high-frequency artifacts in fake images. The discriminator network uses spectral residual analysis to detect frequency components associated with manipulations. The adversarial loss encourages the generator to minimize detectable frequencies, simulating real-world evasion tactics. The iterative training process enhances the robustness of the detection system against adversarial attacks, making it resilient to manipulated images with suppressed high-frequency cues. The model is evaluated on multiple datasets, showing improved performance against advanced forgery techniques [8].

A New Approach for Effective Medical Deepfake Detection in Medical Images adapts YOLO architectures with custom modifications, such as multi-head attention and residual blocks, to detect synthetic medical images. The models use feature pyramids and spatial attention modules to focus on subtle deformations and texture inconsistencies in clinical images. The training dataset includes large collections of knee X-rays and lung CT scans, augmented for variation. The models are trained using object detection loss functions and evaluated for accuracy, precision, and recall. The results show near-perfect detection accuracy, highlighting the importance of explainable and real-time forensic methods in medical imaging [9].

DeepFake Detection for Human Face Images and Videos: A Survey reviews deepfake detection techniques, focusing on pixel-level inconsistencies, biological signals, temporal mismatches, and artifact frequencies. It analyzes the strengths of deep learning classifiers, including CNNs, RNNs, and hybrid models, in capturing spatial and temporal anomalies. The survey discusses vulnerabilities to adversarial attacks and emphasizes research directions such as explainability, transfer learning, and contextual verification. It provides a comprehensive overview of datasets, evaluation metrics, and challenges in building generalizable deepfake detection systems [10].

DeepfakeUCL: Deepfake Detection via Unsupervised Contrastive Learning introduces an unsupervised contrastive learning framework for deepfake detection. The model generates multiple augmented views of each image using random cropping, color jittering, and geometric transforms. A contrastive loss is used to maximize the agreement between augmented views, learning invariant feature representations without labeled data. The learned embeddings are evaluated on multiple benchmarks, achieving competitive detection accuracy compared to supervised methods. The approach is scalable and label-efficient, making it suitable for low-resource scenarios [11].

Deep Learning-Based Gender Classification by Training With Fake Data explores the use of GAN-generated synthetic face images for training gender classification models. CNNs are trained on artificial faces, learning style-agnostic features based on domain-specific textures and facial geometry. The models are evaluated on real-world datasets, showing high accuracy comparable to models trained on real data. The study demonstrates the potential of synthetic datasets to address privacy issues and data scarcity in biometric applications, and suggests future work in feature visualization and classification of other facial attributes [12].

Dodging DeepFake Detection via Implicit Spatial-Domain Notch Filtering proposes DeepNotch, a neural network that applies learned spatial filters to mask frequency signatures associated with deepfakes. The filter is trained on manipulated and real images to suppress artifact-prone frequency bands while preserving semantic content. Adversarial training with a frequency residual loss minimizes detectable cues, significantly decreasing detection scores for state-of-the-art models. The method is evaluated on multiple GAN-based image generation techniques, demonstrating its effectiveness in evading current deepfake detectors and highlighting the ongoing arms race in forgery detection [13].

Voice–Face Homogeneity Tells Deepfake introduces a multimodal detection strategy that compares biometric consistency between an audio track and the corresponding facial video. The approach models voice–face homogeneity: natural speech and facial motion exhibit correlated attributes that synthetic manipulations often fail to reproduce. By extracting cross-modal embeddings and measuring coherence, the system detects mismatches indicative of audio-visual forgeries. Experiments on audio–visual forgery datasets demonstrate that leveraging multimodal correlations significantly boosts detection accuracy, especially for sophisticated manipulations that separately preserve unimodal realism. The study advocates multimodal consistency as a robust cue for next-generation deepfake detection [14].

Exploiting Visual Artifacts to Expose Deepfakes and Face Manipulations systematically analyzes common visual artifacts—such as blending boundaries, color mismatches, and local inconsistencies—left by face manipulation pipelines. The authors design feature descriptors and classifiers that explicitly target typical artifact patterns across spatial scales. By combining handcrafted and learned cues, the method achieves reliable detection across varied manipulation techniques. The paper documents practical artifact types and their forensic relevance, providing both empirical evidence and detection baselines. This foundational work frames many subsequent detector designs by emphasizing the forensic value of artifact-aware feature engineering in exposing manipulated facial imagery [15].

MRE-Net: Multi-Rate Excitation Network for Deepfake Video Detection proposes a multi-rate excitation architecture that processes temporal signals at different frame rates to capture forgery artifacts occurring at varied temporal frequencies. The network employs rate-specific feature extraction branches and a fusion mechanism to integrate complementary temporal cues. This design enhances sensitivity to brief, high-frequency anomalies as well as slower temporal inconsistencies. Evaluations show improved detection on video datasets containing diverse manipulation lengths and frame-rate variations. The paper demonstrates that multi-rate temporal modeling can better represent the heterogeneous temporal signatures introduced by modern face-synthesis techniques [16].

AVoiD-DF: Audio–Visual Joint Learning for Detecting Deepfake Videos develops a joint audio–visual learning framework that combines modality-specific detectors with cross-modal interaction modules. The method learns to exploit temporal alignments and semantic correlations between speech and facial motion to detect inconsistencies introduced by manipulation.

By training jointly on audiovisual data, AVoiD-DF demonstrates robustness to manipulations that preserve unimodal realism but distort cross-modal coherence. Experiments indicate superior performance on datasets containing dubbed or re-synchronized content. The study highlights the importance of integrating audio and visual streams to capture forgery cues that single-modality detectors miss [17].

FaceForensics++: Learning to Detect Manipulated Facial Images introduces FaceForensics++, a comprehensive dataset and benchmark for face manipulation detection, alongside baseline detection architectures. The dataset contains multiple manipulation methods (face swapping, facial reenactment, and more) applied to real videos, enabling systematic evaluation. The paper provides standard evaluation protocols and demonstrates that while deep models can detect many manipulations, performance degrades under compression and cross-dataset tests. FaceForensics++ has since become a canonical resource for training and evaluating forgery detectors, catalyzing research into robustness and generalization for practical forensic deployment [18].

Quality-Centric Framework for Generic Deepfake Detection reframes detection as a quality-assessment problem, proposing a framework that models image and video quality degradation patterns introduced by synthesis pipelines. Instead of relying on manipulation-specific artifacts, the method learns generalized quality discrepancies between real and synthetic content, using hierarchical quality-aware features and attention mechanisms. This approach improves robustness to unseen generation techniques and post-processing operations like compression. Experimental results show competitive cross-dataset performance, advocating quality-centric representations as a complementary pathway toward universal deepfake detectors adaptable to diverse, real-world distributions [19].

Protecting World Leaders Against Deep Fakes addresses a practical and security-critical application: safeguarding public figures from malicious deepfakes. The authors propose detection strategies tailored to high-value targets, combining robust watermarking, detection pipelines, and human-in-the-loop verification. The work includes empirical analysis showing that targeted protections (e.g., proactive media-signing, rapid detection alerts) can reduce the risk of misinformation propagated by manipulated video. The paper advocates coordinated technical and policy-level defenses to protect civic trust, stressing that detection alone is insufficient without provenance and rapid response mechanisms for high-impact deepfake incidents [20].

III Survey Methodology

Inspired by the standardized evaluation pipelines evident in recent research (see Figure 1), this survey adopted a systematic multi-stage methodology for reviewing deepfake detection literature. The process begins with the **collection of input data**, where published benchmarks or datasets (such as FaceForensics++, DFDC, or Celeb-DF) are uploaded and checked for format and resolution compatibility. Only samples passing validity checks (regarding features or metadata) are retained for downstream analysis; invalid entries are filtered out and logged for transparency, mirroring common best practices in contemporary research.

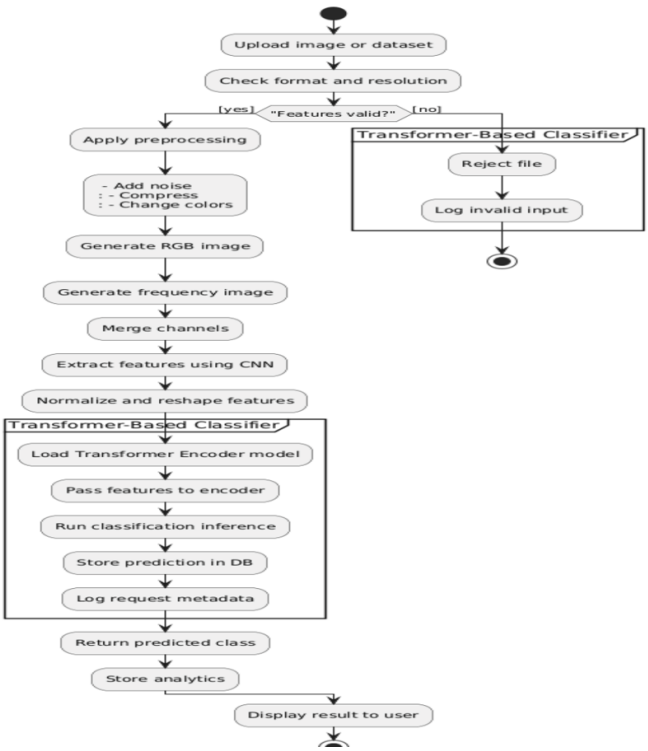


Fig. 1. Flowchart illustrating the survey methodology for deepfake detection studies.

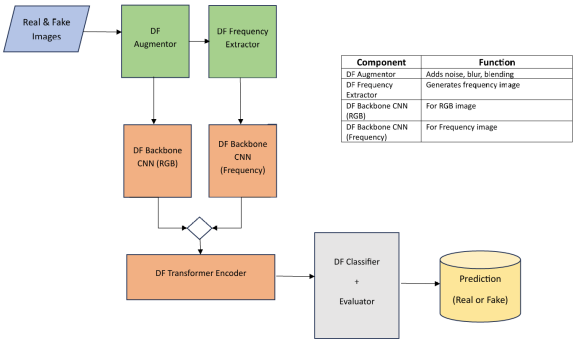
Next, selected images and videos undergo a **preprocessing** phase to enhance robustness and standardization. Common transformations, as depicted in the workflow, include noise injection, compression, or color adjustment—designed to simulate real-world variation and minimize bias. Both RGB and frequency-domain representations are generated, often via Fourier or wavelet transforms, enabling studies to compare and fuse pixel-space with spectral features.

Feature extraction proceeds via convolutional neural networks (CNNs), which are widely employed to harvest spatial features such as facial textures, boundaries, and blended regions. After normalization and reshaping, features are passed into advanced models—most notably, transformer-based classifiers. The transformer component processes spatial and (when applicable) temporal relationships, using multi-head self-attention to deliver nuanced classification decisions. The resulting predictions, feature embeddings, and model metadata are systematically recorded, supporting reproducibility and analytic evaluation.

Finally, predicted classes, analytics, and metadata are compiled, facilitating a holistic review of detection accuracy, model generalization, computational efficiency, and real-time

capability. This modular, step-wise methodology is reflected in our survey's structure: grouping prior works by the main architectural choices (CNN, Transformer, frequency analysis, hybrid), and consistently reporting outcomes in terms of accuracy, generalization, efficiency, and scalability. As shown, clear workflow delineation and robust logging at each stage were emphasized in both literature selection and result synthesis, ensuring a comprehensive, transparent, and reproducible survey process.

IV Existing Approaches



Component	Function
DF Augmentor	Adds noise, blur, blending
DF Frequency Extractor	Generates frequency image
DF Backbone CNN (RGB)	For RGB image
DF Backbone CNN (Frequency)	For Frequency image

Fig. 2. Overview of components and workflows in current deepfake detection strategies, highlighting spatial, spectral, and hybrid feature pipelines.

Over the last several years, deepfake detection has progressed from handcrafted forensic analysis and early convolutional classifiers to increasingly sophisticated, multi-modal hybrid frameworks. This section synthesizes the principal technical directions and recurring findings reported across recent surveys and specialized studies, focusing on what is common and broadly observed in the literature without relying on specific citations.

IV-A CNN-Based Approaches

Convolutional Neural Networks (CNNs) remain a cornerstone due to strong spatial feature extraction. Representative CNN detectors have been effective at identifying local visual inconsistencies, texture artifacts, and boundary anomalies introduced by generative models. Across widely used benchmarks such as FaceForensics++ and DFDC, CNN-based systems still provide reliable baselines when domain characteristics remain stable. A recurrent limitation, however, is sensitivity to distribution shifts: performance can degrade under severe compression, with novel generation techniques, or when models are evaluated cross-dataset, indicating reliance on dataset-specific cues rather than intrinsic manipulation patterns.

IV-B Transformer-Based Models

Transformer architectures have reshaped visual forensics by enabling long-range contextual modeling through self-attention. They have proven well-suited for both spatial and temporal anomaly detection, and variants that integrate frequency cues into the backbone achieve fine-grained discrimination that often surpasses classic CNNs under diverse manipulations. Transformer hybrids can also be paired with domain adaptation modules to improve cross-dataset generalization. The primary trade-off is computational cost: memory and latency requirements remain higher than those of compact CNNs, motivating research into efficient variants, pruning, and quantization.

IV-C Frequency-Domain Analysis

Frequency-based techniques exploit the spectral fingerprints left by generative and face-swapping models. Approaches using power spectrum analysis, Fourier- or cosine-domain transforms, and targeted artifact mining can separate real from synthetic content even when pixel-space cues are subtle. These methods typically show improved robustness to common post-processing such as compression and resizing. Nevertheless, as generation pipelines evolve to suppress or mask frequency artifacts, frequency-only detectors can be evaded; in practice, spectral features are most effective when fused with spatial evidence.

IV-D Hybrid Architectures

State-of-the-art systems increasingly adopt hybrid pipelines that unify spatial CNN features, spectral representations, and transformer-based global reasoning. A common pattern is to extract localized textures, fuse them with frequency components, and apply attention mechanisms to model long-range structure and temporal consistency. Data-centric strategies—such as strong augmentations (e.g., self-blending), adversarial training, and domain/style adaptation—further enhance reliability in the wild. The main drawback is complexity: larger parameter counts and multi-branch designs raise inference cost, prompting current work on lightweight hybrids and self-supervised pretraining to balance accuracy and deployability.

Taken together, experience across these directions shows that combining spatial, spectral, and context-aware modules in hybrid architectures yields steady gains, yet persistent challenges remain: generalization to unseen forgeries, efficiency for edge deployment, and robustness against both post-processing and deliberate artifact suppression. A comparative summary can be placed in an appendix table to highlight trade-offs among accuracy, robustness, and computational cost.

V Research Gaps and Challenges

Although significant progress has been made in deepfake detection, several gaps and challenges continue to hinder the development of robust and scalable solutions:

- **Overfitting and Generalization:** Many methods perform well on benchmark datasets but often fail to generalize to unseen forgery types or diverse real-world data. This is mainly because models frequently rely on dataset-specific cues rather than learning deep intrinsic manipulation patterns.
- **Dependence on Labeled Data:** Most approaches require extensive annotated datasets for effective training.

Maintaining and updating such datasets can be resource-intensive, especially as new forms of manipulations quickly emerge. This dependency affects the scalability and adaptability of detection systems.

- **Computational Complexity:** Transformer-based and hybrid models, while effective, typically demand high computational resources for both training and inference. This can hinder deployment in real-time systems, mobile applications, or environments where lightweight models are required.
- **Robustness Under Distortions:** Detection methods may experience significant drops in performance when faced with compression, blur, adversarial noise, or other distortions common in practical scenarios. Since deepfakes are shared widely across platforms with varying data quality, ensuring robust detection under such conditions is essential.
- **Ethical and Societal Concerns:** Technical advancements in deepfake detection also introduce ethical challenges, such as potential misuse for surveillance, censorship, or privacy invasion. Responsible deployment, transparency, and ethical considerations are necessary alongside technical progress.

VI Future Directions

Despite impressive progress in deepfake detection, multiple open challenges remain—particularly in generalization, efficiency, and real-world resilience. Recent literature and benchmark studies illuminate several promising avenues for future research:

1) **Lightweight Hybrid Architectures:** State-of-the-art deepfake detectors often rely on sophisticated hybrid frameworks that unify convolutional neural networks, transformers, and frequency-domain analysis for robust spatial, spectral, and contextual reasoning. However, many of these models are large and require substantial computational power. Emerging research is focused on developing lightweight variants using model pruning, quantization, or efficient neural architecture search. The goal is to maintain high accuracy while enabling scalable, real-time deployment on edge devices and mobile platforms, thus bridging the gap between laboratory benchmarks and practical use.
2) **Self-Supervised and Few-Shot Learning:** A persistent challenge is the dependence on vast, well-labeled datasets, which limits the adaptability of current detection systems. New approaches—such as contrastive learning without explicit labels, sophisticated data augmentation techniques like self-blending, and rapid adaptation through meta-learning—can help models to recognize new manipulation types by extracting useful patterns from unlabeled or minimally labeled data. These techniques aim to make deepfake detectors more agile in responding to evolving threats and novel attack styles.
3) **Cross-Domain Generalization:** Many detection systems experience a marked decline in accuracy when confronted with unfamiliar datasets, manipulations, or content types. Recent advances in domain adaptation and transfer learning have introduced strategies that align or recalibrate model features to bridge gaps between different data distributions. This enables robust performance across varied sources—an essential requirement as deepfakes spread across international and cross-platform digital environments.
4) **Robust Training Strategies:** Simply training models with standard datasets is not sufficient for real-world robustness. Incorporating training samples that include a variety of distortions—such as lossy compression, blurring, and adversarial noise—can help models learn to withstand the degradations common in practical scenarios. Additionally, leveraging both spatial and frequency-domain artifact cues ensures greater resilience to sophisticated attacks. Ongoing research also emphasizes data-centric enhancements like domain-specific augmentation and adversarial training.
5) **Ethical Frameworks and Standards:** As deepfake detection moves from the lab to broad societal deployment, ethical considerations come to the forefront. Collaborative efforts involving researchers, policymakers, and industry are essential to establish transparent benchmarks, standardized reporting, and evaluative protocols. Issues such as privacy, fairness, explainability, and secure deployment need to guide not only the design of new detection algorithms but also their real-world implementation.

By focusing on these research frontiers, the community can accelerate progress toward scalable, trustworthy, and ethically responsible deepfake detection systems suited for diverse real-world applications.

VII Conclusion

Deepfake detection technology continues to be a pivotal area of research aimed at preserving digital trust, privacy, and societal well-being. Advances in CNN-based methods, Transformer architectures, and frequency-domain analysis have substantially enhanced detection accuracy. In particular, hybrid frameworks that synergistically combine spatial, temporal, and frequency features currently represent the state-of-the-art in both robustness and generalization. Despite these gains, significant challenges remain, including overfitting to limited datasets, high computational demands, and vulnerability to common real-world distortions such as video compression, noise, and adversarial attacks.

Future directions should focus on developing *lightweight yet effective* hybrid models that balance accuracy with computational efficiency to support deployment in resource-constrained and real-time environments. Self-supervised and unsupervised learning paradigms offer promising avenues for reducing labeled data dependency, enabling greater adaptability across diverse manipulation techniques and domains. Domain adaptation techniques and robust feature extraction remain critical to improving cross-dataset generalization.

Additionally, ethical considerations surrounding fairness, privacy, and transparent deployment must be integrated into evaluation frameworks to ensure responsible use of detection technologies. By addressing these intertwined technical and societal challenges, the research community can foster the development of scalable, trustworthy, and robust deepfake detection systems essential for mitigating the pervasive risks posed by synthetic media in real-world applications.

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